

attribute, the attribute value SHALL NOT change. If the fan is in a state where this attribute cannot be changed to the requested value, the server SHALL return INVALID_IN_STATE.

When this attribute is successfully written, the server SHALL set the value of the [FanMode](#) and [SpeedSetting](#) (if present) attributes to values that abide by the mapping requirements listed below.

4.4.6.3.1. Percent Rules

It is up to the server implementation to map between ranges of the [PercentSetting](#) attribute, the [SpeedSetting](#) (if present) attribute, and the [FanMode](#) attribute enumerated values that are supported as indicated in the [FanModeSequence](#) attribute per the requirements below.

There SHALL be a mapping between the values of the [FanMode](#) attribute and the values of this attribute by dividing the domain of values of this attribute into ranges as follows:

- The ranges SHALL be non-overlapping, contiguous, and in ascending order.
- Every value from 0 to 100 SHALL be in exactly 1 range.
- Every value in a lower range shall be less than every value in any higher range.
- The value 0 (zero) in this attribute SHALL be in a range by itself. This range SHALL map to Off.
- If Auto is supported by the server, the null value SHALL be in a range by itself. This range SHALL map to Auto.
- There shall be a one-to-one mapping between each range and each value in [FanMode](#).
- Every range SHALL map to a greater value in the [FanMode](#) attribute than all lower-valued ranges.

The following table illustrates an example of such a mapping. The actual mapping provided by the server is determined by the manufacturer subject to the rules mandated in this specification. Note that the values in this example are contiguous, non-overlapping and span from 0 to 100 with no missing values. Note also that each range appears in an ascending order, 0 appears by itself in a range that maps to Off, and null appears by itself in a range that maps to Auto.

Attribute	Values				
FanMode	Off	Low	Medium	High	Auto
PercentSetting	0	1 to 33	34 to 66	67 to 100	null

If a client successfully sets the [FanMode](#) attribute, the server SHALL set the [PercentSetting](#) attribute to some value within the range which maps to the [FanMode](#) setting. If no such value exists, the attribute value SHALL NOT change. If a client successfully sets the [PercentSetting](#) attribute, the server SHALL set the [FanMode](#) attribute to the value of the corresponding range, and the [SpeedSetting](#) attribute (if present) according to the equation below. If a client successfully sets the [PercentSetting](#) attribute to 0, the server SHALL set the [FanMode](#) attribute to Off and the [SpeedSetting](#) (if present) attribute to 0.

If the [MultiSpeed](#) feature is supported, the value of the [SpeedSetting](#) and [SpeedCurrent](#) (speed) attributes SHALL be calculated from the [PercentSetting](#) or [PercentCurrent](#) (percent) attributes respectively using the following formula: